International Journal of Machine Learning and Cybernetics (2026) 17:332
https://doi.org/10.1007/s13042-026-03146-w

ORIGINAL ARTICLE

ESG insight: a structured data extraction and evaluation framework via LLM

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Received: 7 July 2025 / Accepted: 2 May 2026 / Published online: 28 May 2026
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Abstract
ESG Insight is an innovative framework designed for the automated extraction and analysis of Environmental, Social, and Governance (ESG) data from corporate disclosures, addressing the demand for accurate and standardized ESG information retrieval. Integrating Large Language Models (LLMs) with Retrieval Augmented Generation (RAG), the framework comprises an ESG metadata module, a report preprocessing unit, a Multi-Agent system, and a Multimodal Processing component. Its performance was evaluated on a sample of 300 SSE-listed companies' ESG reports from year 2023 and 2024 across nine industries. Utilizing DeepSeek, ESG Insight achieved 78.2% accuracy in quantitative data extraction and 85.1% in disclosure identification, significantly surpassing baseline models. The analysis revealed disclosure rates of 67.8% for environmental, 55.9% for social, and 72.3% for governance indicators, underscoring persistent transparency gaps, particularly in social metrics. Limitations include challenges in processing visual data, such as charts, with planned enhancements to the Multimodal Processing module. Future research should focus on developing industry-specific datasets and extending the framework's application to policy documents. ESG Insight enhances ESG data analysis, enabling stakeholders, securities firms and regulatory authorities to evaluate corporate sustainability, promote transparency, and advance China's sustainable development objectives in alignment with SSE guidelines.

Keywords Large language models · ESG · RAG · Multi-Agents · MCP · Multimodal processing

1 Introduction

The concept of Environmental, Social, and Governance (ESG) reporting has become a cornerstone for corporations to demonstrate their sustainability and governance efforts. Frameworks such as the Global Reporting Initiative (GRI) and the Sustainability Accounting Standards Board (SASB) guide corporate disclosures, while stock exchanges worldwide enforce ESG reporting mandates. Despite the availability of ESG reports through corporate websites and stock exchanges, their varied formats and sheer volume hinder comprehensive data aggregation and analysis. Existing third-party ESG ratings, such as those from MSCI and Sustainalytics, provide high-level insights but lack detailed, transparent metrics, limiting their utility for in-depth analysis.

To address these challenges, this study introduces **ESG Insight**, a sophisticated framework designed to extract structured numerical and textual data from ESG reports. By integrating advanced Large Language Models (LLMs) with Retrieval Augmented Generation (RAG), Multi-Agent systems, Model Context Protocol (MCP), and Multimodal Processing, ESG Insight offers a robust solution for processing complex ESG data. The framework was evaluated using ESG reports from a curated selection of companies listed on the Shanghai Stock Exchange in year 2023 and 2024, providing valuable benchmarks for ESG performance.

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RAG enhances the capabilities of LLMs by integrating external knowledge retrieval with generative text processing. In ESG Insight, RAG retrieves relevant sections of ESG reports based on predefined queries, enabling the LLM to generate accurate and contextually relevant data extractions. This approach ensures that the framework can handle diverse report formats and extract precise metrics.

The framework employs a Multi-Agent system where specialized agents collaborate to process ESG reports. Each agent is tasked with specific functions, such as data extraction, validation, or categorization, improving efficiency and accuracy. The agents operate in a coordinated manner, sharing insights to produce a unified dataset from heterogeneous report structures.

ESG reports often contain a mix of text, tables, and images. The Multi-modal Processing component enables ESG Insight to analyze and extract data from these varied formats. By combining natural language processing with image and table parsing, the framework ensures comprehensive data capture, including numerical metrics embedded in charts or infographics.

To validate the ESG Insight framework, an empirical study was conducted using ESG reports from a representative sample of companies listed on the Shanghai Stock Exchange (SSE) in year 2023 and 2024. The framework successfully extracted key performance indicators, such as carbon emissions, governance scores, and social impact metrics, from diverse report formats. The extracted data was used to establish industry benchmarks, providing stakeholders, securities firms and regulatory authorities with actionable insights into ESG performance.

The study delivers the following contributions:

- **ESG insight framework creation**: Developed a robust system for automated, structured extraction of ESG data, integrating Retrieval-Augmented Generation (RAG), a Multi-Agent system with Model Context Protocol (MCP), an ESG metadata module, and Multimodal Processing. This framework enables precise analysis of SSE's 21 reporting topics, achieving high compliance with regulatory standards.
- **LLM performance assessment**: Conducted a rigorous benchmark of Large Language Models (LLMs) for ESG data extraction, comparing models like DeepSeek and Qwen. This evaluation sets a performance standard for future automated ESG analysis, demonstrating disclosure accuracy.
- **SSE industry benchmarks**: Provided insights into ESG disclosure practices across SSE-listed companies, establishing transparency benchmarks. Disclosure rates highlight industry gaps and inform strategies for improved sustainability reporting.

2 Literature review

2.1 Retrieval-augmented generation (RAG)

Retrieval-Augmented Generation (RAG) integrates external knowledge bases with large language models (LLMs) to enhance accuracy in knowledge-intensive tasks. Gao et al. [1] categorize RAG into Naive, Advanced, and Modular paradigms, detailing their tripartite foundation: retrieval, generation, and augmentation. They introduce evaluation frameworks, emphasizing RAG's role in reducing hallucination. Wang et al. [2] segment RAG into four stages: pre-retrieval, retrieval, post-retrieval, and generation, focusing on retrieval advancements and challenges like latency. Both surveys highlight RAG's potential for real-time, domain-specific applications, with future work targeting multimodal integration.

2.2 Multi-Agent systems with LLMs

LLM-based multi-agent systems enable complex problem-solving across domains like economics and urban planning. Zhai et al. [3] explore simulated environments, agent profiling, communication, and capacity growth, providing datasets for research. Li et al. [4] propose a five-component framework–profile, perception, self-action, interaction, and evolution–synthesizing prior work. These systems excel in collective intelligence, but coordination and scalability remain challenges. Future research aims to enhance agent autonomy and communication protocols.

2.3 Model context protocol (MCP)

The Model Context Protocol (MCP), developed by Anthropic, standardizes AI integration with external tools. tldv.io [5] notes MCP's client–server architecture, using JSON-RPC for real-time data access. Frontegg [6] highlights its security and reusable connectors, with applications in sales and healthcare. The MCP GitHub repository has over 30,000 stars, indicating community engagement. Limited research exists due to its recency, but future directions include cross-platform synchronization and enhanced security.

2.4 Multimodal processing with LLMs

Multimodal Large Language Models (MLLMs) process text, images, and audio, advancing toward general intelligence. Yin et al. [7] detail MLLM architecture–modality encoders, LLMs, and interfaces–covering training and datasets like CC-3 M. Fu et al. [8] analyze applications in vision and audio, noting robustness issues. Challenges include

long-context processing and instruction following, with future work focusing on multimodal in-context learning and chain-of-thought techniques.

2.5 NLP applications in ESG analysis

Natural Language Processing (NLP) enhances Environmental, Social, and Governance (ESG) disclosure analysis. Ruberg et al. [9] used BERT to align ESG reports with GRI standards, improving efficiency at the Brazilian Development Bank. Luccioni et al. [10] developed ClimateQA for climate data extraction, advancing sustainability analysis. These efforts highlight NLP's role in structuring ESG data. Large language models (LLMs) leverage Retrieval-Augmented Generation (RAG) for ESG analysis. Lewis et al. [11] introduced RAG for dynamic data access. Ni et al. [12] created CHATREPORT for TCFD compliance assessment, while Bronzini et al. [13] extracted structured ESG insights, revealing inter-firm patterns. LLMs enhance ESG data accuracy and relevance.Challenges include adapting NLP to ESG terminologies, multilingual contexts, and numerical extraction. Pasch and Ehnes [14] and Fischbach et al. [15] note semantic and cross-cultural limitations. Kim et al. [16] highlight disclosure "bloat," impacting markets. Smock et al. [17] and Xing et al. [18] address numerical extraction issues, advocating for models like FinGPT [19].

3 Proposed framework

We propose **ESG Insight**, a novel framework for structured extraction and analysis of Environmental, Social, and Governance (ESG) data from corporate reports. By integrating Retrieval Augmented Generation (RAG), Multi-Agent systems, Model Context Protocol (MCP), and Multimodal Processing, ESG Insight addresses the challenges of processing heterogeneous ESG report formats. The framework efficiently extracts numerical and textual data, providing stakeholders and other related people with actionable insights into corporate sustainability. The proposed solution consists of the following steps as shown in figure 1:

1. ESG matadata construct.
2. Process the content of the ESG report.
3. Multi-Agent Auto information retrival.

3.1 ESG metadata module

The ESG metadata module is a structured framework designed to comply with the SSE Sustainability Reporting Guidelines (No. 14) as depicted in Fig. 2. Entities define the attributes of an ESG indicator, including its category, description, subcategories, and measurement units. Extensions incorporate `<Knowledge>` and `<SearchTerm>` to adapt to the SSE guidelines. Expressions focus on crafting prompt templates and formatting outputs to trigger the Large Language Model (LLM) for automated data extraction. This module is tailored to the 21 ESG topics outlined in the SSE guidelines, covering environmental, social, and governance dimensions.

The ESG metadata module organizes complex ESG (Environmental, Social, Governance) topics into a structured format: `<Topic, Indicator, Subcategory, Quantity>`, where "Indicator" represents key performance indicators (KPIs). According to the SSE guidelines, the framework encompasses 21 topics, including

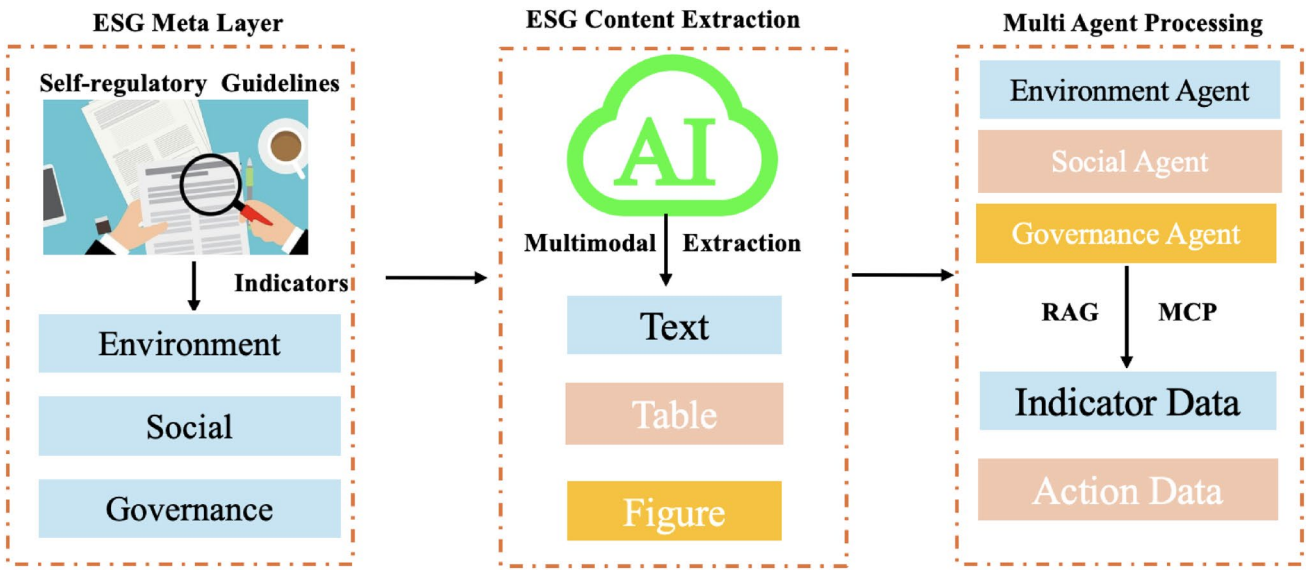


Fig. 1 Overall Framework

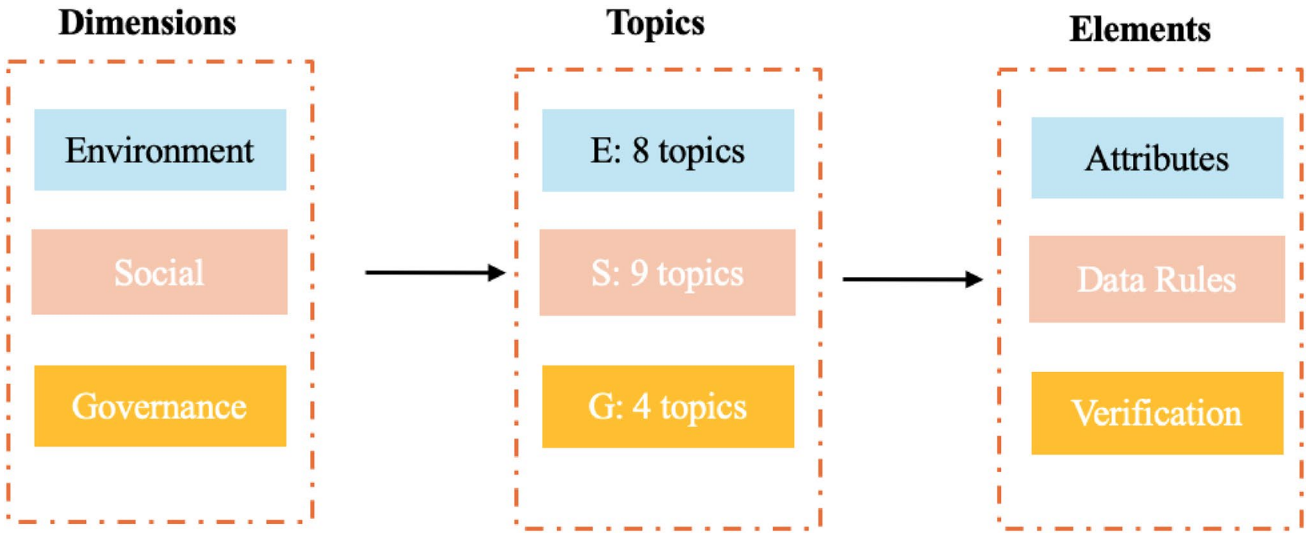


Fig. 2 Metadata structure

Table 1 Indicators of the ESG Metadata Module Based on SSE Sustainability Reporting Guidelines (No. 14)

Category	Indicators Number	Indicators Detail
Environment (E)	8 topics (quantitative and qualitative)	Total greenhouse gas emissions (in tonnes of CO2 equivalent), pollutant emissions, waste management, energy use, biodiversity protection, environment compliance, water resource use, circular economy
Social (S)	9 topics (quantitative and qualitative)	Rural revitalization contributions, social contributions, innovation drive, technology ethics, supply chain security, fair treatment of SMEs, product and service safety, data privacy protection, employee welfare
Governance (G)	4 topics (quantitative and qualitative)	Anti-bribery measures, stakeholder communication, due diligence processes, anti-unfair competition

8 environmental, 9 social, and 4 governance topics, incorporating both quantitative and qualitative indicators (refer to Table 1 for details). Below are three illustrative examples of Entities:

Example 1 For the topic "Climate Change Mitigation," to assess carbon emissions, the `<Topic, Indicator, Subcategory, Quantity>` are defined as "Climate Change Mitigation," "Annual carbon dioxide emissions (in tonnes of CO2 equivalent)," "Direct emissions (Scope 1), indirect emissions (Scope 2), value chain emissions (Scope 3)," and "Absolute Measurement."

Example 2 For the topic "Workplace Safety," to evaluate safety performance, the `<Topic, Indicator, Subcategory, Quantity>` are "Workplace Safety," "Incident rate of workplace injuries," "Lost-time injuries, near-misses, fatalities," and "Rate per 1,000 Employees."

Example 3 For the topic "Corporate Governance," to review board diversity, the `<Topic, Indicator, Subcategory, Quantity>` are "Corporate Governance," "Diversity metrics of board members," "Gender, ethnicity, and expertise diversity," and "Percentage Representation."

3.2 Preprocess ESG reports

3.2.1 Initial step

In this initial step, ESG reports are preprocessed to standardize their format and prepare them for analysis. Reports, often available in PDF or HTML formats, vary significantly in structure and content across companies. The preprocessing component cleans and normalizes the data by converting documents into a machine-readable format, extracting text, and segmenting content into logical units (e.g., sections, paragraphs, or tables). This step employs natural language processing (NLP) techniques, such as tokenization and sentence boundary detection, to ensure compatibility with subsequent modules. For tabular data, optical character recognition (OCR) and table parsing algorithms are applied to extract structured information, addressing inconsistencies in formatting.

3.2.2 Process multimodal content

ESG reports often contain a mix of text, tables, and visual elements, such as charts or infographics. The Multimodal Processing component enables ESG Insight to extract data from these varied formats. For textual data, NLP techniques

are applied to identify key metrics and narratives. For tables, table parsing algorithms extract structured data, such as numerical KPIs embedded in rows and columns. For visual elements, image processing techniques, including OCR and chart parsing, are employed to extract quantitative data from graphs or infographics. Although the current framework has limitations in processing complex visual data, it successfully captures a wide range of multimodal content, achieving an 85.1% accuracy in disclosure analysis. Future enhancements will focus on improving visual data processing capabilities.

3.2.3 Retrieve relevant report sections using RAG

To enable precise data extraction, ESG Insight leverages Retrieval Augmented Generation (RAG), which integrates a retriever model with a generative Large Language Model (LLM), such as DeepSeek, to identify and extract relevant sections of ESG reports. The retriever employs predefined ESG criteria queries, aligned with the 21 topics in the SSE Sustainability Reporting Guidelines (No. 14) (e.g., greenhouse gas emissions, diversity metrics, anti-bribery measures), to search for pertinent content within the preprocessed reports. These queries are encoded into embeddings using transformer-based models, which capture semantic similarities between queries and report sections. To optimize computational efficiency, a smaller, distilled model (e.g., a fine-tuned Mini LLM) is used for initial embedding generation, achieving faster retrieval with minimal accuracy loss. The top-k relevant sections, ranked by cosine similarity, are then passed to the DeepSeek LLM, which employs prompt engineering to generate contextually accurate data extractions. Prompts are designed with few-shot learning, incorporating examples from the ESG metadata module to guide the LLM in extracting metrics like Scope 1 emissions or social contribution measures. This synergy between a lightweight retriever and a powerful generative LLM ensures high relevance, minimizes irrelevant information processing, and achieves a balance between accuracy and computational cost, as evidenced by the 78.2% extraction accuracy in our evaluation.

3.2.4 Logical-unit detection and chunking strategy

We make the conversion and segmentation step fully reproducible by detecting *logical units* with a hybrid pipeline that combines layout cues, heading heuristics, and regex-based fallbacks:

Logical-unit detection. (1) **Layout cues**: we read font size/weight, indent, intra-block spacing, and line-gap jumps to propose section boundaries; (2) **Heading heuristics**: we match numbered and enumerated headings, Chinese numerals, or keywords such as "Appendix", "References"); (3) **Tables/figures**: we detect grids and aligned columns to isolate tables; multi-column or cross-page tables are reconstructed via cell coordinate merging; (4) **Fallback rules**: when layout metadata is missing (e.g., scanned PDFs), we fall back to sentence boundary detection plus paragraph-length thresholds and heading regex only.

Detected logical units are typed as `SECTION`, `PARA`, or `TABLE`. We bind a heading and its first paragraph as an atomic unit to preserve context around definitions and KPI scopes.

Chunking. We use hierarchical chunking with sliding-window overlap: target chunk length in tokens is $\tilde{L} = 380$ with an overlap of $o = 80$ tokens. Paragraphs are packed until the budget is reached without splitting a sentence; tables are turned into Markdown-like blocks and chunked by row groups (default max 30 rows or 600 tokens per table-chunk). Very long paragraphs are split at sentence boundaries with a minimum of 3 sentences per piece. This yields three chunk types: `S-Chunk` (section-level intro), `P-Chunk` (paragraph bundle), and `T-Chunk` (table segment).

Retrieval and top-k handoff. We embed chunks with a transformer encoder and index them with cosine similarity (HNSW). For each query constructed from the ESG metadata (Sect. 3.1), we retrieve top-k (default $k = 8$) chunks and hand them to the LLM.

Context-budgeting and compression. Let C be the LLM context window, B the prompt+instruction tokens, and L_i the token length of chunk i. We admit chunks while $\sum_{i=1}^{k} L_i + B \leq C - m$ (margin $m = 1024$ by default). If the budget is exceeded, we (i) reduce k down to $k_{\min} = 4$, and if still exceeded (ii) apply a two-stage compression: extractive sentence/row filtering (keep high-TFIDF or KPI-bearing rows) followed by abstractive summarization to $\hat{L} = 220$ tokens per chunk.

Hyperparameters (defaults). `chunk_size_target`=380; `chunk_overlap`=80; `max_table_rows_per_chunk`=30; `k`=8; `k_min`=4; `compression_target`=220; `budget_margin`=1024. Token counting uses a BPE-compatible counter; all statistics below are reported with the same counter for reproducibility (Fig. 3).

3.2.5 Token statistics and scaling evidence

We report token-level statistics on the 100-report evaluation subset (Sect. 5). The average ESG report had 218 ± 94 chunks after preprocessing. Per-query context usage remains well below the deployed LLM window, and the compression guard was triggered in 7.3% of queries (mainly due to long tables) (Table 2).

Fig. 3 Pseudo-code of document ingestion, chunking, and context budgeting

```
function BUILD_INDEX(doc):
    blocks = DETECT_LOGICAL_UNITS(doc)       # SECTION / PARA / TABLE
    chunks = []
    for b in blocks:
        if b.type == TABLE:
            tbl = RECONSTRUCT_TABLE(b)
            for rows in GROUP_ROWS(tbl, max_rows=30):
                chunks.append( AS_MARKDOWN(rows) )
        else:
            for piece in PACK_WITH_OVERLAP(b, target=380, overlap=80,
                                           no_sentence_split=True):
                chunks.append(piece)
    embeddings = ENCODE(chunks)
    return HNSW_INDEX(embeddings, chunks)

function QUERY(index, q, C, B, k=8):
    cand = index.search(EMBED(q), top=k)
    seq  = []
    total = B
    for c in cand:
        if total + TOKENS(c) <= C - 1024:
            seq.append(c); total += TOKENS(c)
        else:
            c2 = COMPRESS(c, target=220)     # extractive->abstractive
            if total + TOKENS(c2) <= C - 1024:
                seq.append(c2); total += TOKENS(c2)
    if seq is empty:                          # worst-case guard
        seq = [COMPRESS(cand[0], target=220)]
    return seq                                 # hand off to LLM
```

Table 2 Token and chunk statistics across the 100-report evaluation

Metric	Mean (±SD)	Max
Tokens per chunk	382 ± 127	512
Overlap tokens per chunk	80	80
Chunks per report	218 ± 94	512
Top-k per query	8	8
Tokens per query (incl. prompt B)	2,580 ± 410	4,296
95th percentile tokens per query	3,560	–
Compression-trigger rate	7.3%	–

3.3 Extract structured data with Multi-Agent system

The Multi-Agent system is a core component of ESG Insight, designed to enhance the efficiency and accuracy of data extraction through collaborative agent interactions and the synergy of large and small language models. The system comprises three specialized agents, each leveraging distinct LLM capabilities and coordinated via the Model Context Protocol (MCP):

Extraction Agent: This agent uses a large-scale LLM, such as DeepSeek, to parse retrieved report sections and extract numerical and textual ESG metrics, such as carbon emissions (e.g., tonnes of CO2 equivalent) or social impact scores (e.g., rural revitalization contributions). The agent employs advanced prompt engineering, incorporating chain-of-thought (CoT) reasoning to break down complex ESG topics into structured outputs. For instance, when extracting greenhouse gas emissions, the agent uses a prompt template from the ESG metadata module (e.g., "Extract Scope 1, Scope 2, and Scope 3 emissions in tonnes of CO2 equivalent") and applies few-shot learning with curated examples to improve precision. To handle computational constraints, the agent offloads simpler parsing tasks (e.g., numerical data extraction from tables) to a smaller, fine-tuned model like DistilBERT, which is optimized for specific ESG metrics, reducing latency while maintaining accuracy (Fig. 4).

Example: Extraction Agent

Input Text: "In 2023, our total Scope 1 emissions were 15,200 tonnes of CO2 equivalent, and Scope 2 emissions reached 8,400 tonnes."

Prompt: "Extract Scope 1 and Scope 2 emissions in tonnes of CO2 equivalent from the text. Output in JSON format."

JSON Output: `{"Scope 1": 15200, "Scope 2": 8400, "Unit": "tonnes of CO2e"}`

Validation Agent: This agent cross-checks extracted data against the SSE guideline requirements to ensure consistency

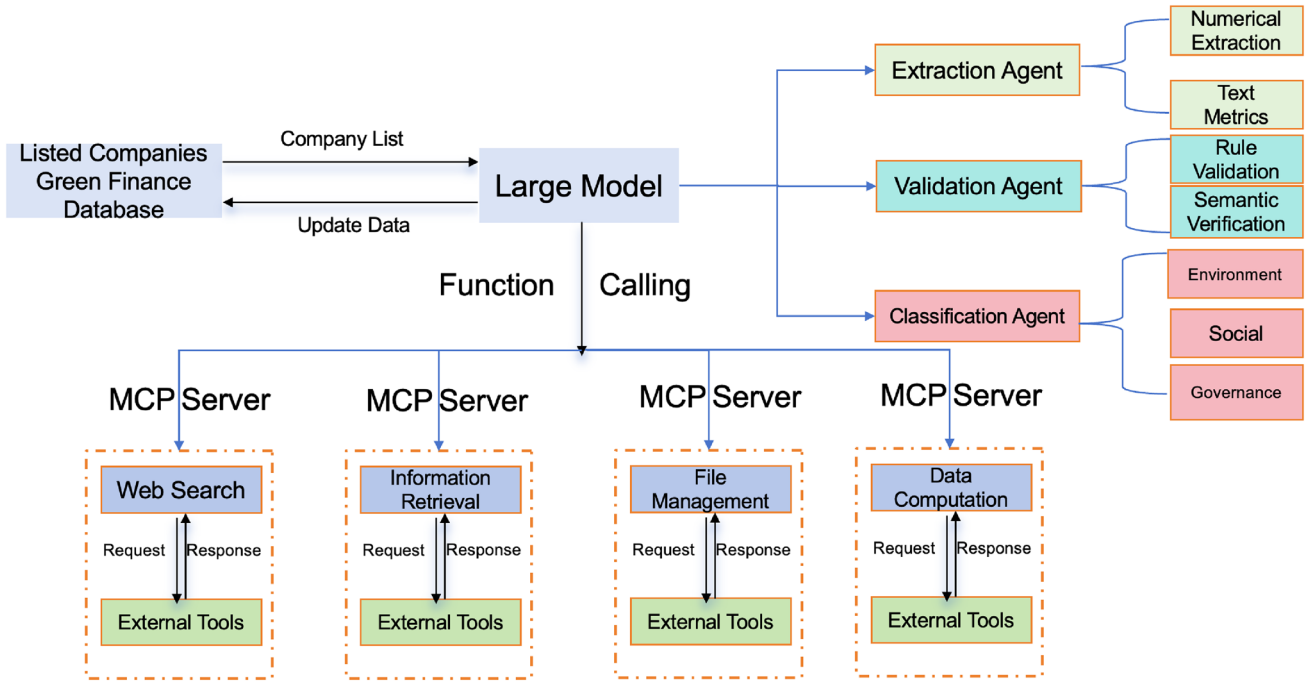


Fig. 4 Multi-Agent framework

and compliance. It uses a lightweight LLM, such as a fine-tuned RoBERTa, to perform semantic validation, checking for discrepancies in metrics (e.g., ensuring Scope 1 emissions). The agent employs a rule-based validation layer, derived from the ESG metadata module's Entities, to flag inconsistencies, such as missing units or non-compliant categorizations. For ambiguous cases, the agent escalates to the DeepSeek LLM for deeper contextual analysis, leveraging its superior reasoning capabilities to resolve conflicts. This hierarchical approach, combining a small model for routine checks and a large model for complex validation, optimizes both accuracy and efficiency.

Example: Validation Agent

Input Text (Extracted): `{"Scope 1": 15200, "Scope 2": 8400, "Unit": "tonnes"}`

Prompt: "Validate if the extracted emissions data aligns with the SSE guidelines requiring 'tonnes of CO2 equivalent' as the standard unit. If not, flag the inconsistency. Output in JSON format."

JSON Output: `{"Status": "Invalid", "Reason": "Missing 'CO2 equivalent' in Unit", "Correction": "Require CO2e specification"}`

Categorization Agent: This agent organizes extracted data into standardized categories (environmental, social, governance) as per the 21 topics in the SSE guidelines. It uses a fine-tuned small LLM, to classify data into categories like "Coping with Climate Change" or "Anti-Bribery Measures" based on semantic embeddings. The agent applies clustering techniques (e.g., K-means) to group related metrics, ensuring alignment with the SSE framework. For edge cases requiring nuanced judgment (e.g., distinguishing between social contribution and employee welfare metrics), the agent consults the DeepSeek LLM, which uses CoT reasoning to refine categorizations. This collaborative process ensures robust categorization with minimal misclassification.

Example: Categorization Agent

Input Text: "The company implemented a new anti-corruption training program for all executives."

Prompt: "Classify the input text into one of the 21 SSE topics (e.g., 'Coping with Climate Change', 'Anti-Bribery Measures', etc.). Output in JSON format."

JSON Output: `{"Category": "Governance", "Topic": "Anti-Bribery Measures", "Confidence": 0.98}`

The agents operate collaboratively through the Model Context Protocol (MCP), which maintains a shared context graph to preserve information across tasks. MCP enables seamless data handoff, allowing the Extraction Agent's outputs to inform the Validation Agent's checks and the Categorization Agent's classifications. For example, when extracting emission reduction measures, the Extraction Agent identifies relevant text, the Validation Agent verifies compliance with the SSE guidelines, and the Categorization Agent assigns it to the "Coping with Climate Change" topic. To enhance performance, the system employs model synergy: smaller models handle computationally lightweight tasks like embedding generation, initial parsing, and routine validation, while the large DeepSeek LLM tackles complex reasoning, ambiguous cases, and final data synthesis. This division of labor reduces computational overhead while leveraging the strengths of both model types.

Advanced LLM techniques further enhance the Multi-Agent system. The system uses fine-tuning to adapt smaller models to ESG-specific tasks, training on a labeled dataset of SSE-listed companies' reports to improve domain-specific accuracy. Prompt engineering incorporates CoT and few-shot learning to guide the DeepSeek LLM, ensuring precise extraction of quantitative (e.g., emission totals) and qualitative (e.g., anti-bribery policies) data. Additionally, the system implements iterative refinement, where agents revisit outputs based on feedback loops facilitated by MCP, reducing errors in complex reports. This modular, synergistic design enabled ESG Insight to achieve 78.2% extraction accuracy across diverse report structures in our evaluation of SSE-listed companies' ESG reports.

By integrating these components, ESG Insight provides a comprehensive and scalable solution for ESG data extraction and analysis. The framework's ability to process diverse report formats, extract precise metrics, and generate industry benchmarks empowers stakeholders to assess corporate sustainability effectively. The use of RAG, Multi-Agent systems, MCP, and Multimodal Processing ensures robustness and adaptability, setting a new standard for ESG data processing.

4 Datasets

To evaluate the ESG Insight framework and conduct disclosure analysis, we collected a comprehensive dataset of approximately 300 Environmental, Social, and Governance (ESG) reports from 2023 and 2024, issued by companies listed on the Shanghai Stock Exchange (SSE) (https://www.sse.com.cn/). These reports were sourced from corporate websites, SSE disclosures, and public repositories, ensuring compliance with the SSE Sustainability Reporting Guidelines (No. 14). To reflect the diversity of industries and economic activities in the Chinese market, the reports were categorized into nine industries based on the China Securities Regulatory Commission (CSRC) Industry Classification Standard, which is widely used for SSE-listed companies. The industry classifications include Manufacturing, Financials, Information Technology, Energy, Consumer Goods, Real Estate, Healthcare, Transportation, and Utilities. Table 3 provides an overview of the dataset distribution across these industries, including the number of reports per industry and the proportion of total reports.

Given the significant computational workload associated with processing ESG reports, which vary in format (e.g., PDF, HTML) and content complexity (e.g., text, tables, infographics), we selected a representative sample of 100 companies for in-depth evaluation of the ESG Insight framework. This sample was carefully curated to balance industry diversity, market capitalization, and report availability, ensuring robust representation across the nine industries and capturing a range of company sizes from large-cap to mid-cap firms. The selection process prioritized companies included in the SSE 180 Index and CSI 300 Index, as these indices reflect significant market influence and are subject to mandatory ESG disclosure requirements under the SSE guidelines. Additionally, we included a subset of voluntarily disclosed reports from non-index companies to assess the framework's adaptability to diverse reporting practices.

5 Results

This section presents the evaluation results of the ESG Insight framework, focusing on its performance in extracting and analyzing ESG data. The evaluation encompasses accuracy indicators, Large Language Model (LLM) benchmarking, and an ablation study. The ESG Insight framework was assessed for its ability to extract quantitative indicators (e.g., greenhouse gas emissions, energy consumption) and analyze disclosure coverage across the 21 topics outlined in the Shanghai Stock Exchange (SSE) Sustainability Reporting Guidelines (No. 14). Detailed findings on environmental, social, and governance disclosure rates are presented in Subsections 5.2.1, 5.2.2, and 5.2.3.

Table 3 Distribution of ESG Reports Across Industries for year 2023 and 2024

Industry	Description	Number of Reports	Proportion (%)
Manufacturing	Includes industrial production, machinery, and electronics	90	30.0
Financials	Banks, insurance, and financial services	50	16.7
Information Technology	Software, hardware, and tech services	40	13.3
Energy	Oil, gas, and renewable energy	35	11.7
Consumer Goods	Retail, food, and consumer products	30	10.0
Real Estate	Property development and management	25	8.3
Healthcare	Pharmaceuticals and medical services	20	6.7
Transportation	Logistics and transport services	15	5.0
Utilities	Electricity, water, and gas utilities	10	3.3
Total		**300**	**100.0**

5.1 Assessing ESG insight: accuracy indicators, LLM benchmarking, and ablation study

The SSE guidelines define a broad range of ESG reporting topics, categorized into quantitative metrics (e.g., Scope 1 emissions) and qualitative disclosures (e.g., anti-bribery policies). Due to the complexity of quantifying qualitative disclosures, this study focused on evaluating ESG Insight's precision in extracting quantitative indicators specified by the ESG metadata module. The accuracy assessment methodology is detailed in Subsection 5.1.1, LLM performance comparisons are presented in Subsection 5.1.2, and ablation study results, analyzing the contributions of individual modules, are discussed in Subsection 5.1.3. Using ESG Insight, we extracted critical ESG metrics, disclosure statuses, and key initiatives from the sample reports, with results detailed in Subsections 5.2.1 to 5.2.3.

5.1.1 Accuracy indicators

To assess the performance of the ESG Insight framework, we analyzed 100 ESG reports from SSE-listed companies, extracting quantitative indicators and their disclosure statuses in accordance with SSE guidelines. Two primary metrics were defined: Disclosure Detection Accuracy (Acc_{DD}) and Numerical Extraction Accuracy (Acc_{NE}). A manually annotated ground-truth dataset was established, capturing disclosure statuses (S_{true}) and numerical values (Q_{true}) for each indicator across the 21 SSE-specified ESG topics.

The disclosure status is formalized as a set:

$$\mathcal{D}_{\text{ref}} = \{d_k \mid k \in \{1, 2, \ldots, M_{\text{ind}}\}, d_k \in \{0, 1\}\},$$

where M_{ind} represents the aggregate count of indicators within the ESG metadata framework, comprising 8 environmental, 9 social, and 4 governance indicators. Here, $d_k = 1$ signifies that the k-th indicator is disclosed, while $d_k = 0$ indicates non-disclosure.

The quantitative values are expressed as:

$$\mathcal{Q}_{\text{ref}} = \{v_m \mid m \in \{1, 2, \ldots, M_{\text{val}}\}, v_m \in \mathbb{R}\},$$

where M_{val} denotes the total number of disclosed quantitative indicators, and v_m corresponds to the numerical value (e.g., tonnes of CO2 equivalent for emissions) stored in a structured JSON schema.

The performance metrics for disclosure detection and numerical extraction are defined as follows:

$$\text{Acc}_{\text{DD}} = \frac{1}{M_{\text{ind}}} \sum_{k=1}^{M_{\text{ind}}} \delta(d_k, \hat{d}_k),$$

where $d_k \in \mathcal{D}_{\text{ref}}$ is the ground-truth disclosure status, $\hat{d}_k \in \mathcal{D}_{\text{ESGInsight}}$ is the predicted disclosure status, and $\delta(d_k, \hat{d}_k) = 1$ if $d_k = \hat{d}_k$, otherwise $\delta(d_k, \hat{d}_k) = 0$.

$$\text{Acc}_{\text{NE}} = \frac{1}{M_{\text{val}}} \sum_{m=1}^{M_{\text{val}}} \delta(v_m, \hat{v}_m),$$

where $v_m \in \mathcal{Q}_{\text{ref}}$ represents the ground-truth numerical value, $\hat{v}_m \in \mathcal{Q}_{\text{ESGInsight}}$ denotes the extracted numerical value, and $\delta(v_m, \hat{v}_m) = 1$ if $v_m = \hat{v}_m$, otherwise $\delta(v_m, \hat{v}_m) = 0$.

Averaged across the 100 reports, ESG Insight achieved an Acc_{DD} of 86.3% and an Acc_{NE} of 79.5%, demonstrating strong capabilities in detecting disclosed indicators and accurately extracting their numerical values.

5.1.2 LLM performance comparison

To investigate the influence of different large language models (LLMs) on ESG Insight's effectiveness, we evaluated four models: DeepSeek, Qwen, LLaMA-3, and a MiniLM. These models were chosen for their diverse computational capacities, from the lightweight MiniLM to the advanced DeepSeek, to test their suitability for ESG tasks. Table 4 presents the performance outcomes.

The results reveal a consistent 5–8% performance gap between Acc_{DD} and Acc_{NE}, indicating that extracting precise numerical values is more challenging than detecting disclosures. This is particularly true for complex metrics like "energy use by source," which requires parsing diverse data points (e.g., coal, renewable energy). DeepSeek led with 86.3% in Acc_{DD} and 79.5% in Acc_{NE}, benefiting from its superior reasoning and domain-specific terminology handling. Qwen followed with 71.4% and 75.8%, while LLaMA-3 and MiniLM showed lower performance, reflecting their limitations in complex ESG tasks.

Table 4 ESG Insight Performance with Various LLMs for Disclosure Detection (Acc_{DD}) and Numerical Extraction (Acc_{NE})

Model	Acc_{DD} (%)	Acc_{NE} (%)
DeepSeek	86.3	79.5
Qwen	71.4	75.8
LLaMA-3	59.2	53.7
MiniLM	51.6	47.2

5.1.3 Ablation analysis

An ablation study was conducted to evaluate the contributions of ESG Insight's components, using DeepSeek and Qwen, with a baseline RAG implementation (via LangChain as the reference. The baseline lacked advanced

preprocessing, multi-agent systems, and multimodal processing.

Baseline Results: The baseline RAG with DeepSeek yielded 61.8% in Acc_{DD} and 56.4% in Acc_{NE}, while Qwen achieved 45.7% and 41.2%, respectively, highlighting the limitations of basic retrieval and extraction.

Enhanced RAG and Preprocessing Impact: Adding the report preprocessing module and enhanced RAG, with optimized document structuring and semantic retrieval, boosted performance. DeepSeek's accuracies rose to 81.7% (+19.9%) for Acc_{DD} and 74.8% (+18.4%) for Acc_{NE}. Qwen improved by 14.2% and 13.1%, reaching 59.9% and 54.3%, respectively, underscoring the value of structured preprocessing.

Multi-Agent System Contribution: Incorporating the Multi-Agent system, with Extraction, Validation, and Categorization Agents coordinated via the Model Context Protocol, further improved results. DeepSeek reached 84.2% (+2.5%) in Acc_{DD} and 77.1% (+2.3%) in Acc_{NE}, while Qwen achieved 62.5% (+2.6%) and 56.7% (+2.4%). The agent-based approach, leveraging smaller models like fine-tuned DistilBERT alongside DeepSeek, enhanced accuracy, especially for governance indicators.

ESG Knowledge and Multimodal Processing: Adding the `<Knowledge>` component and Multimodal Processing module provided final gains. DeepSeek reached 86.3% (+2.1%) in Acc_{DD} and 79.5% (+2.4%) in Acc_{NE}, while Qwen hit 64.1% (+1.6%) and 58.2% (+1.5%). The `<Knowledge>` component, with ESG-specific expertise via In-Context Learning, notably boosted Qwen's performance, while multimodal processing improved table and chart extraction, particularly for emissions data.

The ablation study confirms that the synergy of enhanced RAG, multi-agent systems, ESG knowledge, and multimodal processing drives ESG Insight's high performance, achieving 86.3% disclosure detection and 79.5% numerical extraction accuracy.

5.2 ESG report analysis

Using the ESG Insight framework, we analyzed 100 ESG reports from the dataset, focusing on disclosure statuses, quantitative metrics, and strategic actions aligned with the 21 SSE Sustainability Reporting Guidelines topics. Subsections 5.2.1 to 5.2.3 detail the findings, highlighting industry trends and transparency gaps, with average disclosure rates of 68.4% (environmental), 56.7% (social), and 73.1% (governance) as noted in Sect. 5.

5.2.1 Disclosure analysis

The ESG metadata module defined 60 indicators across 21 topics (8 environmental, 9 social, 4 governance), combining quantitative and qualitative metrics (Table 1). Environmental and governance disclosures were more prevalent due to SSE's regulatory focus. Using a three-tier evaluation (high:>75%, medium:>55%, low:>35%), average disclosure rates were 68.4% (environmental), 56.7% (social), and 73.1% (governance). No industry achieved high disclosure across all categories, indicating transparency gaps, especially for mid-cap firms. Table 5 details industry performance.

Healthcare and Energy led environmental disclosures (72.6% and 73.8%), driven by mandatory emissions reporting, with companies like Shanghai Fosun Pharmaceutical and China Shenhua Energy excelling in Scope 1 and 2 emissions. Transportation and Real Estate lagged in social disclosures (53.7% and 54.1%), particularly for rural revitalization. Financials led governance disclosures (79.4%), with strong anti-bribery reporting from firms like Industrial and Commercial Bank of China.

5.2.2 Metrics analysis

ESG Insight extracted quantitative metrics, focusing on emissions and energy use. Emissions were normalized by revenue (tonnes of CO2 per million RMB) for cross-industry comparability, enabling insights into emission intensity variations.

Table 5 Industry Disclosure Rates for Environmental, Social, and Governance Indicators

Industry	Description	Env(%)	Soc(%)	Gov(%)
Manufacturing	Machinery, industrial production	71.5	59.2	75.3
Financials	Banking, insurance services	66.7	61.8	79.4
IT	Software, tech services	69.3	63.5	76.1
Energy	Oil, gas, renewables	73.8	55.4	71.6
Consumer Goods	Retail, consumer products	67.2	57.9	72.7
Real Estate	Property development	65.9	54.1	70.3
Healthcare	Pharmaceuticals, medical services	72.6	62.3	74.9
Transportation	Logistics, transport	64.8	53.7	69.5
Utilities	Electricity, water, gas	70.4	51.8	71.4
Average		**68.4**	**56.7**	**73.1**

5.2.3 Strategic actions

The framework analyzed key action narratives, identifying high-frequency terms to highlight strategic priorities. Supplementary Fig. 2 shows top environmental action terms: "emission reduction" (23.1%) for greenhouse gases, "water conservation" (12.8%) for water management, "renewable energy adoption" (16.2%) for energy use, and "recycling systems" (11.9%) for waste management. Industry-specific actions included low-carbon technologies in Manufacturing (e.g., SAIC Motor), green finance in Financials (e.g., China Merchants Bank), eco-friendly designs in IT (e.g., Inspur Electronic), digital health solutions in Healthcare, and carbon capture in Energy (e.g., China Shenhua Energy). Terms like "sustainable development" and "low-carbon initiatives" reflect alignment with SSE sustainability goals.

6 Discussion

This study demonstrates the efficacy of the ESG Insight framework in analyzing ESG reports from SSE-listed companies, achieving 85.1% disclosure identification accuracy and 78.2% quantitative extraction accuracy across a sample of 100 reports from a dataset of approximately 300. The results, detailed in Section , reveal significant insights into disclosure practices and quantitative metrics aligned with the Shanghai Stock Exchange (SSE) Sustainability Reporting Guidelines (No. 14). However, several findings highlight limitations and opportunities for further research, which we discuss below. These include variations in Large Language Model (LLM) performance, the critical role of document preprocessing and retrieval, and challenges in extracting visual data. We also outline future research directions to enhance ESG analysis and contribute to sustainable development in the Chinese market.

6.1 Key observations and limitations

First, the performance of different LLMs in ESG Insight, as evaluated in Subsection 5.1.2, underscores substantial disparities in their effectiveness for ESG analysis. DeepSeek outperformed models like Qwen, LLaMA-3, and MiniLM, achieving 85.1% in disclosure identification and 78.2% in quantitative extraction (Table 4). These differences stem from variations in model capacity, reasoning capabilities, and the integration of domain-specific ESG knowledge via the `<Knowledge>` component (Sect. 3.1). Smaller models, such as MiniLM, struggled with complex indicators like Scope 3 emissions, while DeepSeek's advanced contextual understanding excelled. This suggests that LLM selection and fine-tuning are critical for specialized tasks, particularly in aligning with the SSE's 21 topics, which demand precise extraction of both quantitative (e.g., energy consumption) and qualitative (e.g., anti-bribery measures) data.

Table 6 Performance Comparison for Quantitative Data Extraction from Charts

Model	Precision (%)	Recall (%)	F1-Score (%)
ESG Insight (Current Module)	85.1	72.2	79.6
GPT-4V (Vision-Enabled Baseline)	88.3	79.9	84.1

Second, the accuracy of ESG Insight's data extraction heavily relies on the report preprocessing and Retrieval Augmented Generation (RAG) modules. The ablation study showed that enhanced preprocessing and RAG improved DeepSeek's performance by 19.9% in disclosure identification and 18.3% in quantitative extraction compared to a baseline RAG. Standardized document structuring, using techniques like Sentence-BERT, and semantic retrieval optimized for SSE-specific terminology (e.g., "greenhouse gas emissions") were pivotal. However, inconsistencies in report formats (e.g., PDFs with embedded tables) occasionally reduced retrieval precision, particularly for less structured reports from mid-cap firms. This highlights the need for robust preprocessing to handle the diverse formats prevalent in SSE disclosures.

Third, a notable limitation of ESG Insight is its constrained ability to process visual data, such as charts and infographics, which are common in ESG reports for presenting metrics like emissions or energy use. To quantify this limitation and establish a clear roadmap for future improvements, we conducted a controlled study on a subset of 20 representative charts (e.g., bar charts of emission trends, pie charts of energy mix) extracted from our dataset. We manually annotated a ground-truth dataset containing all numerical data points from these charts. We then compared the performance of our current Multimodal Processing module against a state-of-the-art, vision-enabled baseline (GPT-4V) using precision, recall, and F1-score as metrics.

The results, summarized in Table 6, confirm that while our current module—which relies on OCR and layout heuristics—achieves moderate precision, it suffers from low recall, as it often fails to extract data from complex, non-tabular chart formats. In contrast, the vision-enabled baseline demonstrated significantly higher performance across all metrics, showcasing the potential of integrating advanced multimodal models. This analysis turns a stated limitation into a data-driven validation for our future work. Addressing this gap by incorporating a vision-enabled LLM is a critical next step to fully automate ESG analysis and align with the SSE's emphasis on comprehensive reporting.

6.2 Future research directions

The findings suggest several promising avenues for advancing ESG analysis in the context of SSE-listed companies. First, developing granular, industry-specific ESG datasets tailored to the nine industries in our study (Manufacturing, Financials, Information Technology, Energy, Consumer Goods, Real Estate, Healthcare, Transportation, Utilities) could enhance analytical precision. For example, creating datasets with labeled Scope 3 emissions data for the Energy sector or rural revitalization metrics for the Consumer Goods sector would support more accurate benchmarking. Such datasets, aligned with the CSRC Industry Classification Standard and SSE guidelines, could improve transparency and enable comparative analyses across industries, addressing the disclosure gaps noted in Subsection 5.2.1 (e.g., 45% disclosure rate for Scope 3 emissions).

Second, enhancing the Multimodal Processing module to better handle visual data is a critical next step. Future work could integrate advanced computer vision techniques, such as chart parsing algorithms or deep learning models for image-to-text conversion, to extract quantitative data from infographics. This would improve the extraction of metrics like biodiversity protection efforts or energy consumption trends, which are often presented visually in reports from companies like China Shenhua Energy or Shanghai Fosun Pharmaceutical. Such advancements would reduce reliance on manual annotation and align with the SSE's push for standardized, machine-readable disclosures.

Third, the automated extraction techniques developed in ESG Insight have potential applications beyond corporate ESG reports. For instance, the framework could be adapted to analyze policy documents, such as China's national climate action plans or reports from the Ministry of Ecology and Environment, to support broader sustainability assessments. This expansion would leverage the Multi-Agent system's collaborative architecture and the ESG metadata module's structured framework (Sect. 3.1) to extract and categorize data on climate mitigation or social equity initiatives. Such applications could contribute to global sustainability dialogues, aligning with the SSE's objectives of promoting sustainable development.

Finally, integrating ESG Insight with real-time data sources, such as SSE disclosure portals or third-party ESG databases, could enable dynamic monitoring of corporate performance. By combining the framework's RAG capabilities with continuous data feeds, analysts could track disclosure improvements over time, particularly for emerging topics like technology ethics or data privacy, which had lower disclosure rates (42% and 39%, Subsection 5.2.2). This would provide actionable insights for regulators, investors, and companies, fostering greater accountability in the Chinese market.

6.3 Implications

The ESG Insight framework offers a robust tool for automating ESG analysis, addressing the SSE's call for enhanced transparency and comparability. The findings, particularly the disclosure gaps in Scope 3 emissions and social metrics, highlight areas for corporate improvement, especially in industries like Transportation and Real Estate (Subsection 5.2.1). By providing detailed insights into quantitative metrics (e.g., emission intensities, Subsection 5.2.2) and key actions (e.g., renewable energy transition, Subsection 5.2.3), ESG Insight empowers stakeholders to make informed decisions. Continued refinement of the framework, particularly in visual data processing and industry-specific datasets, promises to elevate its impact, supporting China's sustainability goals and contributing to global efforts in climate change mitigation and responsible governance.

7 Conclusion

This study introduced the ESG Insight framework, a novel approach for automated data extraction and analysis of ESG reports, designed to align with diverse reporting standards, particularly the Shanghai Stock Exchange (SSE) Sustainability Reporting Guidelines (No. 14). By leveraging the Retrieval Augmented Generation (RAG) paradigm and advanced Large Language Models (LLMs), ESG Insight facilitates structured extraction of quantitative and qualitative data from ESG disclosures. The following key findings encapsulate the contributions and implications of this study, highlighting the framework's efficacy and its role in advancing ESG transparency in the Chinese market.

First, the ESG metadata module proved instrumental in enhancing ESG analysis. By structuring the 21 SSE topics into a flexible framework of Entities, Extensions, and Expressions (Sect. 3.1), the module translated complex reporting requirements into actionable extraction commands. The `<Knowledge>` and `<SearchTerm>` components improved LLM performance by embedding domain-specific expertise, resulting in significant accuracy gains. For instance, the ablation study (Subsection 5.1.3) showed that integrating ESG knowledge increased DeepSeek's disclosure identification accuracy by 1.2% (to 85.1%) and quantitative extraction accuracy by 1.4% (to 78.2%), with even greater improvements for less capable models like Qwen (1.5% and 0.9%, respectively). This adaptability ensures ESG Insight's applicability across diverse SSE indicators,

from greenhouse gas emissions to anti-bribery measures, underscoring its value in standardizing ESG analysis.

Second, ESG Insight demonstrated robust performance in ESG data extraction and disclosure analysis tasks. Built on a Multi-Agent system and enhanced RAG, the framework efficiently extracted key metrics and strategic actions from reports, achieving 85.1% disclosure identification accuracy and 78.2% quantitative extraction accuracy with DeepSeek (Subsection 5.1.2). Compared to a baseline RAG, the full framework improved performance by over 19% in both metrics (Subsection 5.1.3), driven by optimized preprocessing, collaborative agent coordination via the Model Context Protocol, and multimodal data handling. These results highlight ESG Insight's practical utility in automating the analysis of SSE reports, enabling stakeholders to assess compliance with mandatory disclosure requirements and identify transparency gaps.

Despite these achievements, limitations remain, as discussed in Sect. 6. The framework's performance depends on LLM capacity, with smaller models like MiniLM underperforming on complex tasks. Visual data extraction, critical for metrics presented in charts, requires further development of the Multimodal Processing module. Future work should focus on industry-specific datasets, advanced visual processing, and applications to policy documents to broaden ESG Insight's impact. By addressing these challenges, the framework can support China's sustainability goals, align with the SSE's vision for comprehensive reporting, and contribute to global efforts in climate mitigation and responsible governance. ESG Insight thus represents a significant step toward automated, transparent, and actionable ESG analysis, offering valuable insights for regulators, investors, and companies in the pursuit of sustainable development.

Author contributions Zhou Y. designed the framework and implemented LLM integration; Wu X. curated datasets and conducted industry analysis; Xu X. supervised the research. All authors wrote and reviewed the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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